

# SAP Fiori UI5 Frontend Component & State Architecture

Decoupled Web Component Hierarchy • Observable Reactive Stores • EventBus Driver

## VIEW & FLOORPLAN LAYER (core/component.ts subclasses)

### ShellBarView & Footer

- Branding & Mode Switcher
- Keycloak User Profile Session
- Global Toast notifications

### ListReportView (Floorplan)

- Multi-field FilterBar (Search, Year, Format)
- Paginated Document Table & Sorting
- Selection & Line-Item Action Handlers

### ObjectPageView (Floorplan)

- Sticky Header with KPIs, Edit & Overwrite
- Dual AI Benchmark Cards (Llama & Mistral)
- Interactive RAG Chat with Citation Drawer

### Modal Dialogs Hierarchy

- UploadDialogView (Multi-file pipeline)
- BpmnModalView (Static BPMN XML View/Download)
- OpenApiModalView & BackendSettings

## OBSERVABLE REACTIVE STORES (core/store.ts)

### appStore

- documents[], activeDocument, activeView
- filters (search, format, year, sortField)
- pagination (page, pageSize, totalPages)

### backendStore

- mode: 'rest' | 'mock' (Dynamic Switch)
- serverUrl: '/api/v1'
- healthStatus & latency measurements

### i18nStore

- currentLocale: en, de, fr, es, ro
- Instant reactive UI string updates
- Date & currency localization formatters

### themeStore

- currentTheme: 'sap\_horizon' | 'sap\_horizon\_dark'
- Sets data-sap-theme on html element
- Injects UI5 shadow DOM stylesheets

## COMMUNICATION & BACKEND GATEWAY LAYER (services/backend/)

### BackendGateway Facade

- BackendGateway.dispatchBackend(op, payload)
- BackendGateway.requestBackend(op, payload)
- Emits lifecycle events: REQUEST, SUCCESS, FAILURE
- Routes dynamically to active adapter driver

### RestBackendAdapter Driver

- fetch() HTTP client talking to /api/v1/\*
- Multipart file upload & FormData encoding
- Handles error envelopes & status codes
- Keycloak Bearer token attachment

### MockBackendAdapter Driver

- Client-side in-memory mock document collection
- Instant zero-latency responses for offline demos
- Full BibTeX formatting & client-side search simulation
- Used when backend server is disconnected